Aegis REBOUND

Recovery-aware respiratory digital twin

Algorithm, data and reproducible evidence

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https://github.com/B-Sheikh/Aegis-REBOUND

A working synthetic proof of concept combining personal historical records, sequential state estimation, competing explanations and budgeted observation selection.

RECAP algorithm specification

State and hypotheses

For each component, x = [d, v, r, b] represents an obstruction proxy, its trend, activity-recovery debt, and wearable respiratory-rate bias. There are three hypothesized explanations (ordinary recovery, airway deterioration, sensor drift) crossed with recovery retention rho in {0.40, 0.64, 0.86}. This gives nine filters. These names describe model families, not ground-truth diagnoses.

The discrete step is five minutes. The transition is:

d_next = 0.995*d + v + exposure_drive(mode)
v_next = 0.93*v
r_next = rho*r + 0.028*(steps/5)
b_next = bias_retention(mode)*b

The airway family receives max(PM2.5 - 34, 0)*0.00019; other families do not. The bias retention is 0.98 for sensor drift and 0.70 otherwise. Mode-specific process-noise matrices and observation-noise values are specified directly in recap.py. All coefficients are engineering assumptions, not physiological constants or clinically fitted parameters.

Observation equations subtract personal resting baselines and instantaneous activity terms:

RR - resting_RR - 0.04*(steps/5) = 7.5*d + r + b + noise
HR - resting_HR - 0.33*(steps/5) = 13*d + 2.4*r + 0.12*b + noise
PEF / personal_best = 1 - 0.60*d - 0.006*r + noise

An independently measured RR is modeled without the wearable bias term; it retains the instantaneous activity contribution. EHR attack history sets the initial family priors. EHR baselines affect measurements throughout replay, so static and dynamic information are fused at every update.

Each component uses a linear Kalman update with Joseph-form covariance correction. State clipping introduces a nonlinear approximation; the covariance is not corrected for truncation. Therefore the resulting Gaussian moments are descriptive. Weight updates use log(weight)*0.96 + log-likelihood*0.30 followed by normalization and a 1% uniform reset mixture. This reset prevents an explanation from becoming irrecoverably zero during a regime change. These are hand-specified robustness choices, not an optimally learned switching process. Components do not exchange state as in a full interacting-multiple-model filter.

Forecast and risk mapping

Each component advances 24 steps with no additional activity and current PM2.5 held constant. Future exposure is not read. Mixture means and total variances are combined by the law of total variance. Bands use mean +/-1.645 standard deviations with display support clipping. They are not conformal or clinically calibrated 90% intervals, and mixture skew makes a Gaussian interpretation approximate.

At each horizon, compute a Gaussian marginal probability of PEF fraction below 0.8. Take the maximum geometric mean over consecutive triples. This is a smooth risk index; multiplying marginal probabilities does not correctly integrate a joint correlated crossing event. A scalar logistic mapping fitted on separate calibration patients maps the index to an empirical future-event probability. Prediction includes ongoing low states, and does not establish detection before onset.

Information objective

Candidates are an optional peak-flow ratio observation and an independent respiratory-rate observation. Each candidate defines a Gaussian predictive observation under every component. Nine-point Gauss-Hermite quadrature, weighted by the component mixture, integrates possible observed outcomes.

For each hypothetical outcome, recompute only hypothesis weights using Bayes likelihood. Hold the component future risk indices fixed. Define U as binary entropy of the mixture risk index plus 0.15 times hypothesis entropy. Expected gain is U before minus expected U after. Score equals this gain minus the assumed burden (0.035 nats for PEF, 0.018 for independent RR). Values are rounded for display and selection. A positive highest score may request a simulated measurement; otherwise choose wait. Enforce no more than two optional probes per day and a 24-step cooldown.

The entropy and burden operate in algorithmic units, not clinical utility. Candidate uncertainty from a full covariance update is not propagated through hypothetical future forecasts. This approximation is deliberately documented rather than mislabeled exact Bayesian experimental design. A more exact extension would update each candidate-conditioned state, reforecast and integrate the downstream clinical loss, but would need a validated loss and measurement model.

When a simulated observation is actually supplied, all components receive both state/covariance updates and revised mixture weights. The data generator provides a new current-time observation; future targets remain unavailable. Optional independent RR has independent measurement noise and no wearable-bias term.

What is proposed and how it is tested

The proposed contribution is the exact integration of a recovery-memory hypothesis bank with a limited-budget disagreement-resolution workflow. It is built from existing methods and is not established as a first invention. Tests remove recovery diversity and EHR personalization, compare simpler classifiers, and compare budgeted adaptive observation with a fixed schedule. These experiments can support or reject a benefit for this implementation in its simulator; they cannot validate biological truth or establish a research priority claim.

Model and data card

Intended use

Local competition research demo for adult asthma monitoring. No clinical validation, patient-care use, diagnosis, treatment recommendation, exposure challenge, deliberate exertion, or emergency assessment. The measurement controls reveal synthetic observations only.

Data

rebound/data.py generates one day of five-minute samples per patient. Seed 71 is used for the primary 88-patient study, split 48/16/24. Twelve additional patients use seed 93 and changed parameters for a stress test. Eight held-out public demo streams are bundled; all other records are reproducible from the generator. These are not real EHRs or wearable records and are not Synthea, MIMIC or AAMOS data. No actual patient files are included.

EHR fields include age, asthma duration, prior attack count, personal-best PEF (L/min), resting respiratory rate (breaths/min), and resting HR (beats/min). Age and asthma duration are displayed but not used by RECAP; attack history sets initial priors. The supervised snapshot baseline also uses age. Dynamic records contain sample index, elapsed minutes, RR, HR, five-minute step count, PM2.5 (micrograms/m3) and optional scheduled PEF.

The four invented scenario families are balanced by patient index. All include ordinary walking. Some include increased exposure-driven obstruction, sensor bias or a mixed pattern. Concealed scenario names are never passed to inference. The nonlinear obstruction and activity-debt simulator uses different transition equations from the filter, but shares hand-authored relationships. This limits independence and makes the benchmark an engineering test rather than a clinical simulation.

Hidden states and dense PEF target arrays exist only inside evaluation or synthetic measurement generation. public_patient() removes them. The predictor can run identically on the public dictionary. A dedicated test verifies this and another modifies all future observations without changing an earlier prediction.

Endpoint and metrics

Three consecutive five-minute future observed PEF fractions below 0.8 of personal best within the next 120 minutes. This includes ongoing low states and is not a clinically adjudicated attack. The simulator observes dense future PEF for scoring; normal input PEF is sparse every four hours. No claimed two-hour lead time follows from a two-hour horizon.

Benchmark issues every 30 minutes from t=36 through t=258. Last-hour HR/RR completeness must be at least 75%, latest paired readings no older than 15 minutes, and at least three hours of history must exist. Missing readings skip wearable updates; no future fill occurs. Values outside the invented model support and device artifacts beyond the synthetic scenarios have not been comprehensively validated.

AUROC, Brier, precision/recall at probability 0.5 and confusion counts describe overlapping issue windows. They are not independent events. PEF forecast MAE is measured as a fraction of personal best, not L/min. The paired policy Brier confidence interval resamples whole patients 1,000 times; it does not represent variation across possible simulator designs. Group differences are toy-scenario diagnostics, not demographic fairness evidence.

Calibration and policy comparison

The no_ehr_personalization ablation removes personal resting HR/RR baselines and attack-history priors; it still retains personal-best PEF for units/normalization. Likewise, the wearable_only artifact name means the snapshot sensor/context feature set: its PEF ratio still depends on known personal-best PEF. It is not a pure no-EHR experiment. Labels in the interface use sensor/context wording; conclusions must keep this limitation explicit.

Snapshot classifiers fit on 48 training patients; normalization is training-only. Filters are hand specified, so no claim of learned physiology is made. Each ablation has its own one-input logistic risk mapping fitted on 16 calibration patients. The frozen passive mapping is reused for probe policies; post-probe calibration may differ and is a limitation. All final results use 24 test patients. Model changes during development are not a preregistered study; the fixed population is not a protected external clinical test.

All policies receive regular PEF every 48 samples. Fixed optional PEF probes occur at t=108 and 204 (09:00 and 17:00), avoiding normal measurement timestamps. Adaptive policy evaluates candidate information on the same 30-minute issue grid, caps optional probes at two and uses a two-hour cooldown. It can use fewer probes; the budget is a common maximum, not necessarily equal realized measurement count. Optional measurements are generated only at their request times. Fixed/adaptive outcomes share the same synthetic patients and target noise.

Uncertainty and safety boundaries

Model-family weights are not probabilities of clinical diagnoses. Recovery debt is a mathematical memory of observed activity, not an established airway biomarker. Forecast bands are Gaussian-mixture moment approximations and have no validated 90% coverage. The risk index is only empirically calibrated on synthetic data. The planner may reduce model ambiguity without improving future-event prediction. Burden weights are assumed, not measured patient preferences.

No raw PPG, cough audio, pulse oximetry, real inhaler integration, validated PM sensor, personal spirometry device or hardware was connected. No India-specific training, prospective silent study, medication model, fairness analysis, regulatory assessment or production security validation was completed.

Related work and proposed contribution

Reviewed 18 October 2026. This is a targeted public-literature check, not an exhaustive systematic review, patent search, or proof that nobody has implemented the same composition.

Related respiratory-twin research

1. Home monitoring in asthma: towards digital twins (2023). https://pubmed.ncbi.nlm.nih.gov/37102597/ . Asthma home-monitoring twins are an established direction. REBOUND focuses on competing recovery explanations and observation selection within that direction.
2. AirPredict (2025). https://www.frontiersin.org/journals/digital-health/articles/10.3389/fdgth.2025.1573342/full . Wearables, personal pollution exposure, spirometry and clinician interfaces already exist together. REBOUND extends the engineering focus to budgeted ambiguity resolution.
3. Slowing Down of Recovery as a Generic Risk Marker (2016). https://doi.org/10.1097/CCM.0000000000001564 . Recovery slowing has been proposed across episodic disorders, including asthma. This provides context for investigating recovery-aware state estimation.
4. OTTER-A, University of Edinburgh project. https://usher.ed.ac.uk/research/medical-informatics/projects/otter-a . Wearable signatures of recovery and deterioration after an asthma attack are being studied. This is relevant background for recovery-focused monitoring.
5. AAMOS-00 / home monitoring dataset (2023). https://www.nature.com/articles/s41597-023-02241-9 . A public research direction already combines connected asthma monitoring with attack prediction. The article identifies 22 phase-two participants and a longitudinal dataset. REBOUND has not downloaded, trained on or validated against these participants.
6. Digital markers of asthma exacerbations: systematic review (2024). https://pmc.ncbi.nlm.nih.gov/articles/PMC11647917/ . Existing marker evidence is heterogeneous; lung-function evidence is conflicting in the review. A plausible wearable association does not validate REBOUND’s endpoint or its hidden-state explanations.

Existing methodological foundations

Kalman state estimation, Bayesian model averaging, multiple-model estimation, expected information gain, Gaussian quadrature and cost-sensitive sensing are existing methods. RECAP does not claim new mathematical foundations. It is not a full interacting multiple-model estimator because states are not mixed between components. Its approximate information objective does not integrate a validated clinical decision loss.

Narrow contribution being proposed

A patient-conditioned bank explicitly competes between activity-recovery, airway-state and sensor-bias explanations; recovery diversity changes inferred temporal state; the system chooses or defers an extra observation under a two-measurement budget; actual new observations revise the displayed hypotheses. The package makes this workflow executable and tests whether the components help. The targeted search did not establish an identical implementation, but that absence does not establish uniqueness.

Falsifiable hypotheses

- Personal EHR baselines improve future low-PEF risk estimates over a population baseline.
- A bank of recovery time constants improves over one fixed retention setting.
- Adaptive information-based observation selection improves risk prediction relative to two fixed additional PEF observations.
- A drifting sensor can sometimes be separated from an airway-state change by an independent observation.

The included experiments evaluate these hypotheses, including comparisons that do not favor the added mechanism. Information gain, predictive performance and clinical utility are distinct objectives.

Clinical context

WHO asthma fact sheet: https://www.who.int/news-room/fact-sheets/asthma . Asthma is a chronic noncommunicable disease; environmental triggers and monitoring are relevant context. WHO South-East Asia burden report: https://www.who.int/southeastasia/publications/i/item/9789290221494 . These sources establish relevance, not simulator validity. No Indian-population prevalence or efficacy statistic is invented for the pitch.

Measured results and interpretation

Canonical results: artifacts/metrics.json, generated by python -m rebound.benchmark. Primary seed 71. The source includes development changes; this is an exploratory synthetic engineering study, not a preregistered or external test.

Future sustained low-PEF event

904 eligible issue windows from 24 held-out test patients, with 298 positives. Passive RECAP AUROC 0.9433, Brier 0.06432, precision 91.27%, sensitivity 84.23%, at threshold 0.5. Counts: TP 251, FP 24, FN 47, TN 582. The endpoint includes ongoing low states, so these statistics do not establish warning before attack onset. The 120-minute forecast MAE is 0.07237 of personal best, or 7.24 percentage points of personal best.

Ablations

The single-recovery-setting bank has AUROC 0.9442 and Brier 0.06330; recovery diversity does not improve event classification here, although PEF forecast MAE is slightly lower for full RECAP (0.07237 versus 0.07477). Removing personal EHR baselines gives AUROC 0.8007 and Brier 0.16493. This supports personalization within this simulator, which deliberately varies personal baselines; it does not validate general patient benefit.

The snapshot fusion classifier has AUROC 0.9487 and Brier 0.05791. Wearable/context-only classifier has AUROC 0.9478 and Brier 0.05736. Both score better than passive RECAP on Brier. The proposed value of the composition is an inspectable dynamical explanation and observation-selection workflow, not guaranteed predictive superiority.

The RR >22 baseline has AUROC 0.6933, Brier 0.24889, precision 65.27%, sensitivity 52.35%. The threshold is an illustrative comparator, not a clinical alarm threshold recommendation.

Measurement policies

All policies have regular four-hourly PEF. Fixed extra PEF at 09:00 and 17:00 uses 48 optional observations across 24 patients, Brier 0.06218, AUROC 0.94657. Adaptive selection uses 38 optional observations, Brier 0.05933, AUROC 0.94633. Adaptive used 20.8% fewer optional measurements and slightly improved pooled Brier, but AUROC was slightly

lower. Both have a maximum two-probe budget, not equal realized counts.

Paired patient-mean Brier difference, adaptive minus fixed: -0.002823; 1,000-resample patient-bootstrap 95% interval [-0.005359, -0.000164]. Negative favors adaptive. This is evidence for the policy in this one invented population, not real-world burden reduction. A single simulator, chosen burdens and approximate information model can create optimistic or fragile results.

Shift

12 new synthetic patients with seed 93, slower activity recovery, changed exposure sensitivity and larger PEF noise: 452 eligible windows, 169 positives, AUROC 0.90592 and Brier 0.07505. This shows performance degradation, and is not independent clinical validation.

Interpretation

These experiments support a reproducible proof of concept for personal conditioning and budgeted observation selection in the stated simulator. They do not establish clinical attack prediction, sensor-fault diagnosis, or general superiority over the comparison models. The hypothesis bank's purpose is to expose evolving model explanations and measurement tradeoffs; its value beyond this synthetic setting remains to be evaluated.

Digital Twin Challenge 2026 - deliverables

Project: Aegis REBOUND - recovery-aware respiratory digital twin with budgeted observation selection.
Team Aegis: Abhishek Bharathi and Aadithya Ram.
College: Amrita Vishwa Vidyapeetham, Coimbatore. No incubator affiliation.
Public repository: https://github.com/B-Sheikh/Aegis-REBOUND

Required materials

- Title, team, use case, stack and model: repository README.
- Minimum 20-minute explanation and prototype demonstration: 20:30 project walkthrough. Disclosed computer narration; presentation slides and application captures. Chapter metadata.
- Architecture in PDF format: Architecture.pdf.
- Presentation with project details and outcomes: Project_Presentation.pdf.
- Open-source license: MIT License.
- Historical plus dynamic data fusion: synthetic generator, RECAP implementation, eight demo streams, and model/data card.
- Working prediction and clinician UI: launch guide, local server, and dashboard.

Supporting evidence

- Technical report, algorithm specification, and related work.
- Results and comparisons, experiment JSON, raw predictions, and tests.
- 15-minute silent product demo, demo chapters, and editable source captures/captions. This shorter edition supplements the complete 20:30 walkthrough.
- Design rationale and release checksums.

All project records are synthetic. Videos and PDFs are included directly in the public repository; no account or additional file permission is required to download them. The distribution ZIP uses the folder name Team Aegis_Amrita Vishwa Vidyapeetham Coimbatore.